

Week 7: Learning Theories - Study Notes

Synthesizing Illeris, Ertmer & Newby, Allen, Greeno et al., and multimedia sources

Article Summaries

Illeris (2018): A Comprehensive Understanding of Human Learning

Illeris provides a sweeping historical overview of learning theory from pre-1950 to contemporary times, organizing the evolution around his comprehensive three-dimensional model of learning: content (knowledge and skills), incentive (emotion, motivation, and volition), and interaction (social and environmental context). His account traces behaviorism's dominance through the mid-20th century, the cognitive revolution of the 1950s-70s, and the emergence of constructivist and social learning theories in the 1980s-90s. Particularly significant is his attention to the 1970s emergence of liberatory and emancipatory approaches including Freire's pedagogy of the oppressed, Negt's work with trade unions, and Mezirow's transformative learning. These critical perspectives recognized that learning is inherently political and that educational systems often reproduce rather than disrupt power structures.

The three-dimensional model itself represents a significant conceptual advance by refusing to separate cognition from emotion and social context, thereby rejecting the Cartesian dualism that has historically justified reductionist educational practices. Illeris explicitly critiques Skinner's behaviorism as being beyond any understanding of human qualities and warns against contemporary neo-liberal competition state education policies that prioritize economic considerations over learning theory and human motivation. He introduces the concept of learning barriers including mislearning, learning defense, and learning resistance, suggesting that what gets labeled as educational failure might sometimes represent rational responses to oppressive pedagogical conditions.

However, the account reveals significant limitations in its scope and critical depth. Illeris explicitly acknowledges his overview is geographically constrained to European, American, and Australian contributions, lacking engagement with Asian and African learning traditions. This reflects the ongoing coloniality of educational psychology, where Western theories get universalized as human nature while other epistemologies remain marginalized. Additionally, while the article grants space to neuroscientific research on learning, including claims about gendered brain differences, it doesn't fully interrogate how biological explanations have historically been weaponized to justify educational inequality based on gender, race, class, and disability.

Despite cataloging critical pedagogy and transformative learning, Illeris treats them as one approach among many rather than as fundamental challenges to the entire enterprise of neutral educational psychology. The comprehensive model, while progressive, still operates within assumptions that all learning can be studied as an

individual psychological process somewhat separate from power relations. The account documents how educational psychology has repeatedly discovered and then forgotten that learning is fundamentally about meaning-making, agency, and transformation, with each paradigm shift claiming to overcome prior reductionism while introducing new forms of control.

Ertmer & Newby (1993/2013): Behaviorism, Cognitivism, Constructivism

Ertmer and Newby's foundational article compares three major learning theories—behaviorism, cognitivism, and constructivism—through seven key questions about how learning occurs, what factors influence it, what gets learned, how transfer happens, what types of learning are best explained, and how instruction should be designed. Their central argument positions these theories not as competing frameworks but as complementary approaches suited to different learning contexts. They propose that instructional designers should select theoretical approaches based on the learner's level of expertise (novice to expert) and the cognitive processing demands of the task (low to high complexity).

Behaviorism focuses on observable behavior changes, viewing learning as association formation strengthened through reinforcement. Knowledge exists outside the learner and is transmitted through carefully designed stimulus-response sequences. Cognitivism shifts attention to mental processes, conceptualizing learning as information processing where the mind actively organizes and stores knowledge in schemas. Knowledge becomes internal mental representations that can be strategically processed. Constructivism goes further, positioning learners as active meaning-makers who construct understanding through experience and reflection. Knowledge is personally constructed and socially mediated, with learning occurring through authentic tasks in rich contexts.

The article's continuum model suggests using behaviorist approaches for novices learning foundational skills, cognitivist strategies for intermediate learners developing expertise, and constructivist methods for advanced learners tackling complex, ill-defined problems. The 2013 update addresses how Web 2.0 technologies enable new forms of constructivist learning through participatory cultures, authentic audiences, and distributed cognition. It acknowledges the rise of connectivism as a potential fourth learning theory for the digital age, though questions whether it represents a distinct epistemology or an extension of existing frameworks.

However, viewing these theories as complementary tools obscures their fundamentally different assumptions about human agency, knowledge, and power. Each theory embodies political commitments about who learners are and what education is for. Behaviorism treats learners as objects to be shaped, cognitivism as processors to be programmed, and constructivism as sense-makers to be scaffolded—yet even constructivism typically positions teachers as experts who control what gets constructed. The framework offers instructional designers flexibility but doesn't interrogate whose interests each approach serves or how educational systems use theory to legitimate existing inequalities.

Allen (2006): Overview and Evolution of the ADDIE Training System

Allen traces the evolution of ADDIE (Analyze, Design, Develop, Implement, Evaluate) from its post-World War II military origins through four distinct generations of development. First generation ADDIE (1950s-1970s) emerged from military training needs, grounded in behaviorist principles of task analysis and measurable objectives. Second generation (1970s-1980s) incorporated cognitive psychology, emphasizing mental models and information processing. Third generation (1980s-2000s) integrated constructivist principles and rapid prototyping methodologies. Fourth generation (2000s-present) adapts to distributed learning environments, performance support systems, and agile development processes.

Despite these theoretical shifts, ADDIE's core structure remains remarkably stable across generations: systematic analysis of learning needs, intentional design of solutions, development of materials, implementation of training, and evaluation of effectiveness. The model is presented as neutral, scientific, and broadly applicable across contexts—from military training to corporate learning to educational settings. Each generation claims to overcome the limitations of prior approaches through more sophisticated theories and technologies, yet the fundamental purpose remains unchanged: efficiently training people to perform predetermined tasks.

The article explicitly acknowledges ADDIE's roots in creating technical training programs for new military recruits who were to function in standardized work roles. This origin story is not incidental but fundamentally shapes ADDIE's epistemological assumptions. The system was designed to produce compliant workers who could perform tasks efficiently, not to develop critical consciousness or agency. This mirrors what critical pedagogy critiques as the banking model of education, where knowledge is deposited into passive learners rather than co-created through dialogue.

ADDIE's evolution demonstrates how instructional design can appropriate progressively novel theories instrumentally—to make training more effective—without taking seriously their broader epistemological and political implications. Truly embracing situative perspectives, for example, would require acknowledging that workplace practices themselves are contested terrains where power operates, and that learning to participate might mean learning to question and transform those practices, not just comply with them. ADDIE excels at what it was designed to do: produce workers who can perform predetermined tasks efficiently. But this kind of education-as-training differs fundamentally from education-as-conscientization.

Greeno, Collins, & Resnick (1996): Cognition and Learning

Greeno, Collins, and Resnick provide a comprehensive analysis of three major theoretical perspectives on cognition and learning: behaviorist/empiricist, cognitive/rationalist, and situative/pragmatist-sociohistoric. Each perspective offers distinct answers to fundamental questions about what it means to know, how learning occurs, how knowledge transfers, and what motivates engagement. The behaviorist view conceptualizes knowing as associations strengthened through practice, learning as stimulus-response conditioning, transfer as extending response patterns to new stimuli, and motivation as extrinsic reinforcement. The cognitive view treats knowing as conceptual understanding and problem-solving strategies, learning

as construction of mental models, transfer as applying general principles to new problems, and motivation as intrinsic interest in intellectual challenge.

The situative perspective represents the most radical departure, conceptualizing knowing as ability to participate in community practices, learning as increasingly central participation in authentic activities, transfer as preparation for future learning in related practices, and motivation as engaged participation in valued communities. This view emphasizes that cognition is distributed across people and tools, situated in specific contexts, and fundamentally social. Knowledge doesn't reside in individuals' heads but in the practices of communities. Learning isn't acquisition of decontextualized skills but transformation of participation in culturally organized activities.

The article explores practical implications for designing learning environments, formulating curricula, and constructing assessments. Behaviorist approaches emphasize information transmission and training environments, component-to-composite skill sequences, and measurement of discrete knowledge components. Cognitive approaches favor problem-solving and exploratory environments, conceptual development sequences, and evaluation of extended performance. Situative approaches design for participation in social practices, organize curricula around disciplinary practices and realistic problems, and assess participation in inquiry practices. The authors discuss exemplary programs including the Algebra Project, Treisman's Professional Development Program, and Eckert's study of high school social groups.

However, presenting these perspectives as complementary tools for different purposes obscures their fundamentally different political commitments. Each embodies assumptions about human agency, legitimate knowledge, and the purpose of education. The situative perspective offers the most promising resources for critical pedagogy by foregrounding how learning is always learning to participate in communities of practice—but this raises urgent questions about which communities, whose practices, and what forms of participation. Well-intentioned communities of practice in schools might inadvertently require students to abandon home languages, cultural ways of knowing, or community epistemologies to be recognized as successful learners. The question isn't which perspective is most accurate, but whose interests each serves.

Video Summaries

Todd Rose: The Myth of Average (TEDxSonomaCounty)

Todd Rose's TED talk challenges the entire paradigm of designing for the average person, using the Air Force's 1950s cockpit crisis as a powerful historical example. When the Air Force measured 4,063 pilots on ten dimensions to design a better average cockpit, they discovered that zero pilots actually fit the average profile across all dimensions. This revelation exposed how designing for average inevitably designs for no one, leading to the innovation of adjustable equipment. Rose extends this insight to education, arguing that standardization based on average profiles systematically fails to serve actual learners, whose multidimensional variability means almost no one is average. The jaggedness principle suggests that

performance is not one-dimensional—a student might excel in creative writing but struggle with grammar, and these dimensions don't correlate predictably.

However, Rose's framing focuses heavily on high-potential dropouts as a lost resource for innovation and economic productivity. While the critique of standardization is valuable, positioning gifted dropouts as the primary concern risks reinforcing meritocratic ideologies that value only certain kinds of difference. A critical perspective asks: What about students whose variability doesn't align with innovation or economic value? The call to design for individuals resonates with critical pedagogy's commitment to honoring learner diversity, yet remains under-theorized about the power relations that determine which differences matter and who benefits from personalization.

The ADDIE Model of Instructional Design (YouTube Educational Video)

This instructional video provides a straightforward overview of ADDIE's five phases: Analyze (identify learning needs and learner characteristics), Design (specify learning objectives and assessment strategies), Develop (create instructional materials and resources), Implement (deliver the instruction), and Evaluate (measure effectiveness and gather feedback). The video presents ADDIE as a systematic, iterative framework that ensures instructional quality through evidence-based design. It emphasizes how each phase builds on previous ones while allowing for revision based on formative evaluation data.

The video's neutral, procedural tone exemplifies how ADDIE gets taught as technical methodology divorced from questions about power, purpose, and values. The framework is presented as applicable to any learning context without interrogating whose needs get analyzed, who defines objectives, which materials get developed, how implementation contexts shape outcomes, or what counts as effectiveness in evaluation. Missing entirely is attention to how systematic design can serve either domestication or liberation depending on who controls the process and for what ends.

Synthesis: Learning Theories as Political Choices

These six sources, viewed together, reveal a fundamental tension in educational theory and practice: the persistent myth that learning theory can be politically neutral, scientifically objective, and universally applicable. From Illeris's comprehensive historical overview through Ertmer and Newby's theoretical comparison, Allen's ADDIE evolution, Greeno's three perspectives, and both multimedia sources, a pattern emerges of progressive refinement within existing paradigms rather than revolutionary transformation of educational purposes. Behaviorism gave way to cognitivism, which yielded to constructivism, which now contends with situativity and connectivism—each claiming to overcome prior limitations while maintaining education's fundamental project of making learning predictable, measurable, and manageable. Yet this entire evolutionary narrative obscures a more fundamental question: Whose interests do these theories serve?

The historical trajectory documented across these sources shows how educational theory has repeatedly discovered and then systematically forgotten that learning is political. Illeris notes the 1970s emergence of liberatory approaches (Freire, Negt, Mezirow) that recognized education either domesticates or liberates, yet his own comprehensive model treats critical pedagogy as one option among many rather than as a fundamental challenge to neutral educational psychology. Ertmer and Newby's continuum suggests selecting theories based on learner expertise and task complexity, positioning theories as complementary tools—yet this instrumental framing erases their incompatible assumptions about human agency, knowledge, and power. Allen's ADDIE evolution demonstrates how instructional design appropriates progressive theories to improve training efficiency without interrogating training's purposes. Greeno's three perspectives acknowledge different epistemologies but present them as serving different design needs rather than embodying different political commitments.

Todd Rose's myth of average critique resonates powerfully with these theoretical tensions. If designing for average systematically fails because no one is actually average across dimensions, then learning theories that position learners as categories (novices needing behaviorist instruction, intermediates needing cognitive scaffolding, experts needing constructivist challenges) similarly fail to serve actual people. Yet Rose's focus on gifted dropouts as lost resources reveals how even critiques of standardization can reinforce meritocratic ideologies. The question isn't just whether we design for average or individual—it's whether we design for compliance or conscientization, for reproduction or transformation, for domestication or liberation. Illeris's three dimensions (content, incentive, interaction) offer more promise by refusing to separate cognition from emotion and social context, yet still position learning as individual psychological process somewhat separate from power relations.

What remains persistently uninterrogated across most of these sources is who defines learning needs, whose knowledge counts as legitimate, which communities' practices get valued, and what purposes education serves. ADDIE's analyze phase asks what learners need to know—but not who decides what counts as need. Ertmer and Newby discuss transfer—but not how transfer presumes that school knowledge is more legitimate than community knowledge. Greeno's situative perspective emphasizes participation in practices—but not how practices themselves reflect and reproduce power structures. Rose critiques average—but not how individuality gets commodified in neoliberal markets. Even Illeris's comprehensive history, while attending to liberatory approaches, maintains geographical constraints that reflect educational theory's ongoing coloniality, where Western frameworks get universalized while Indigenous, African, and Asian epistemologies remain marginalized.

The synthesis that emerges is both sobering and generative. Sobering because it reveals how educational theory has been remarkably effective at absorbing critique—each generation of theory claims to overcome prior reductionism while introducing new forms of control, each calls for learner-centered approaches while maintaining expert control, each champions diversity while perpetuating which differences matter. Yet generative because understanding these patterns enables more strategic resistance. If learning theory cannot be neutral, then the task becomes making political commitments explicit and designing accordingly. If theories embody power

relations, then the work is disrupting those relations through practice. If educational systems were designed to sort rather than develop, to domesticate rather than liberate, then transformation requires not better theories within existing systems but dismantling and reimagining what education is for. These sources provide vocabulary and frameworks for understanding learning, but realizing liberatory education requires asking different questions: Learning for what? Defined by whom? In whose interest? Toward what future?